Preliminary Assignment for Cyber Security

PrestaShop Deployment on Amazon Web Services

Student	Abdulwahab
Platform	AWS Academy Learner Lab
Application	PrestaShop 9.2.0
Date	October 4, 2026

Public storefront URL at time of submission:

http://100.27.194.138/index.php

Note: This deployment uses an AWS Academy lab environment. The public IP may change if the instance is stopped/recreated or the lab environment is reset.

1. Objective and Architecture

The goal of this assignment was to create a new server instance, install PrestaShop, make the store publicly accessible, and keep the database on a separate server. I used an Amazon EC2 instance for the web application and Amazon RDS for the MySQL database. This keeps the web server and database separated, as required by the assignment.

Internet → EC2 Web Server (Apache + PHP + PrestaShop) → Amazon RDS MySQL

Security decisions included restricting SSH to a trusted source, allowing public HTTP/HTTPS access to the web application, keeping RDS non-public, and allowing MySQL traffic on port 3306 only from the PrestaShop security group.

2. EC2 Web Server Creation

I created an Ubuntu EC2 instance named PrestaShop-Web-Server. The instance was placed in the AWS lab VPC and assigned a public IPv4 address so the PrestaShop storefront could be reached from the Internet.

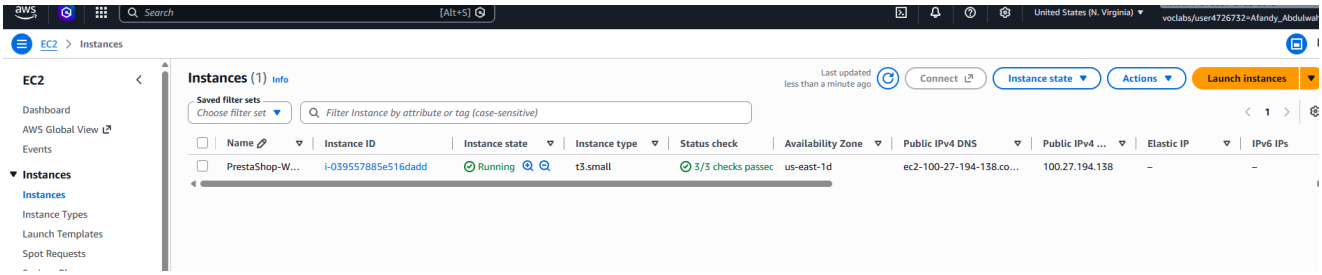


Figure 1. EC2 instance running with a public IPv4 address.

3. Separate RDS MySQL Database

I created an Amazon RDS MySQL database named prestashop-database. The RDS instance was configured in the same VPC as the EC2 server but was not made publicly accessible. This satisfies the requirement that the database must not be hosted on the same server as the application.

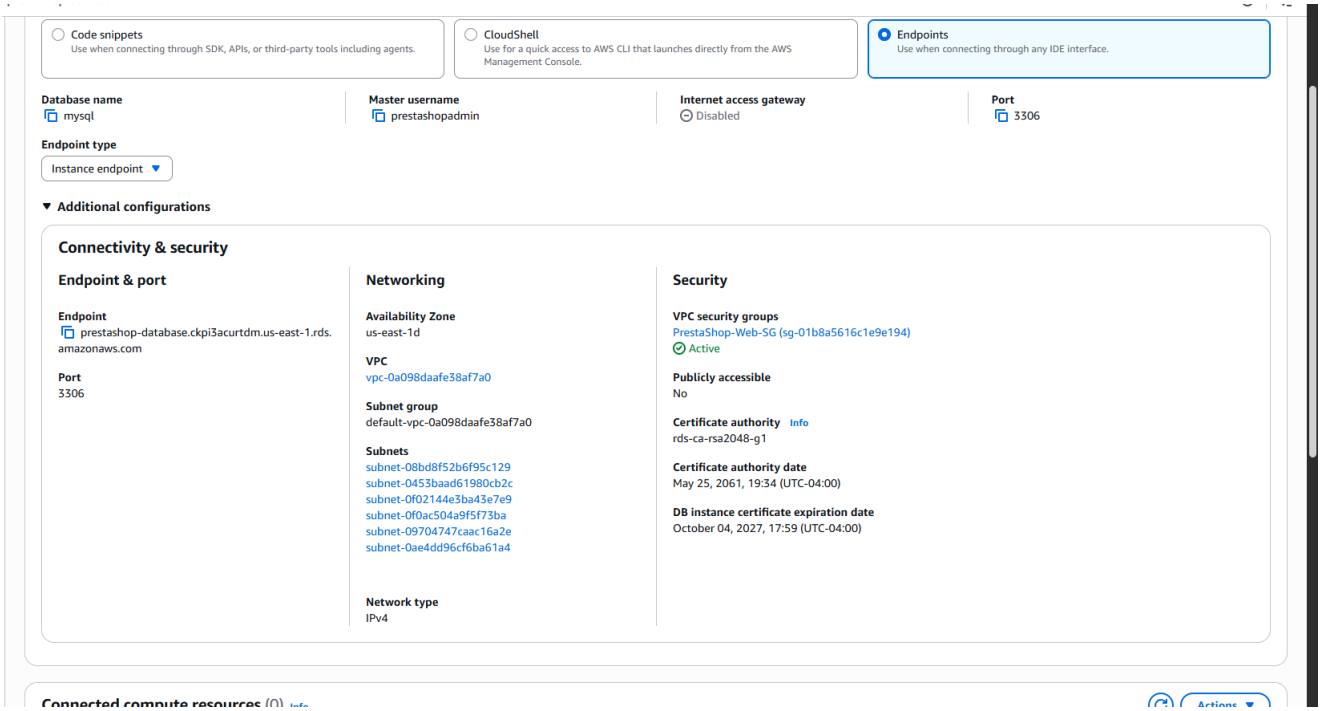


Figure 2. RDS connectivity details showing port 3306 and Publicly accessible: No.

4. Security Group Configuration

The PrestaShop security group was configured with only the network access needed for the deployment. SSH was restricted, HTTP and HTTPS were allowed for public website access, and MySQL/Aurora port 3306 was restricted to the PrestaShop security group instead of being opened to the whole Internet.

Traffic	Port	Access
SSH	22	Restricted source / administration
HTTP	80	0.0.0.0/0 for public storefront
HTTPS	443	0.0.0.0/0 for web traffic
MySQL/Aurora	3306	PrestaShop security group only

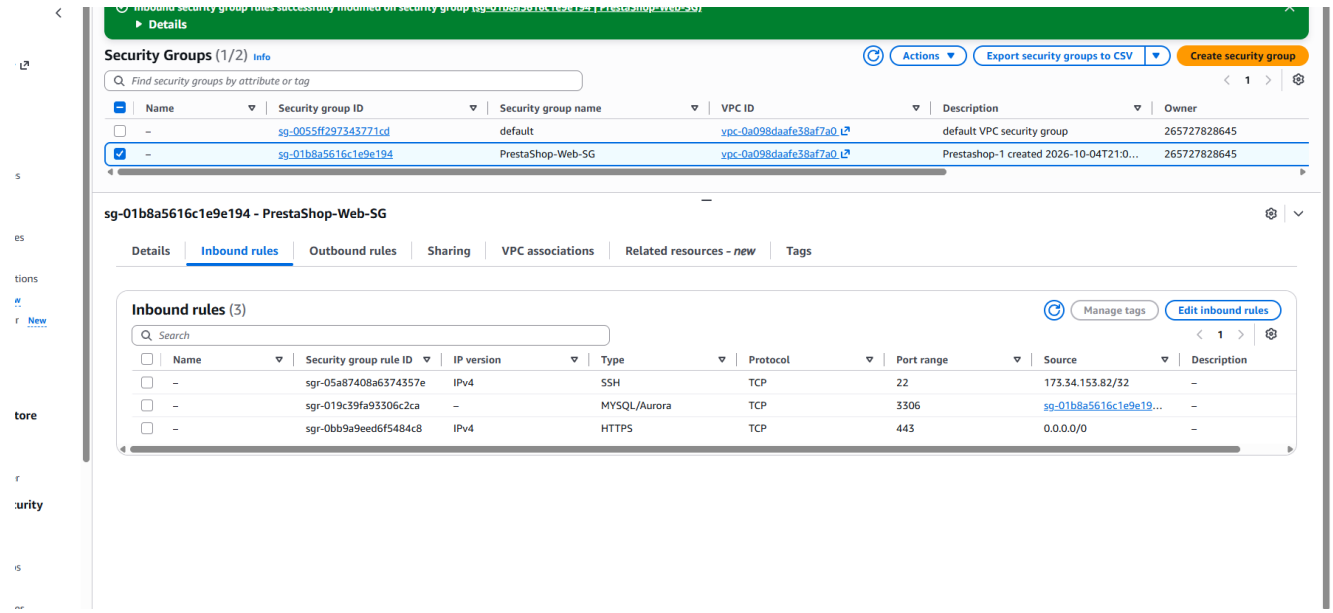


Figure 3. Security group rules, including restricted MySQL access on port 3306.

5. Apache and PHP Installation

I connected to the Ubuntu EC2 instance using SSH and updated the package list. I then installed Apache, PHP, the required PHP extensions, unzip, and other supporting packages. Apache was enabled and started successfully.

```
sudo apt update
sudo apt install -y apache2 php php-cli php-common php-curl php-gd php-intl php-mbstring php-mysql php-xml php-zip unzip wget
sudo systemctl enable apache2
sudo systemctl start apache2
```

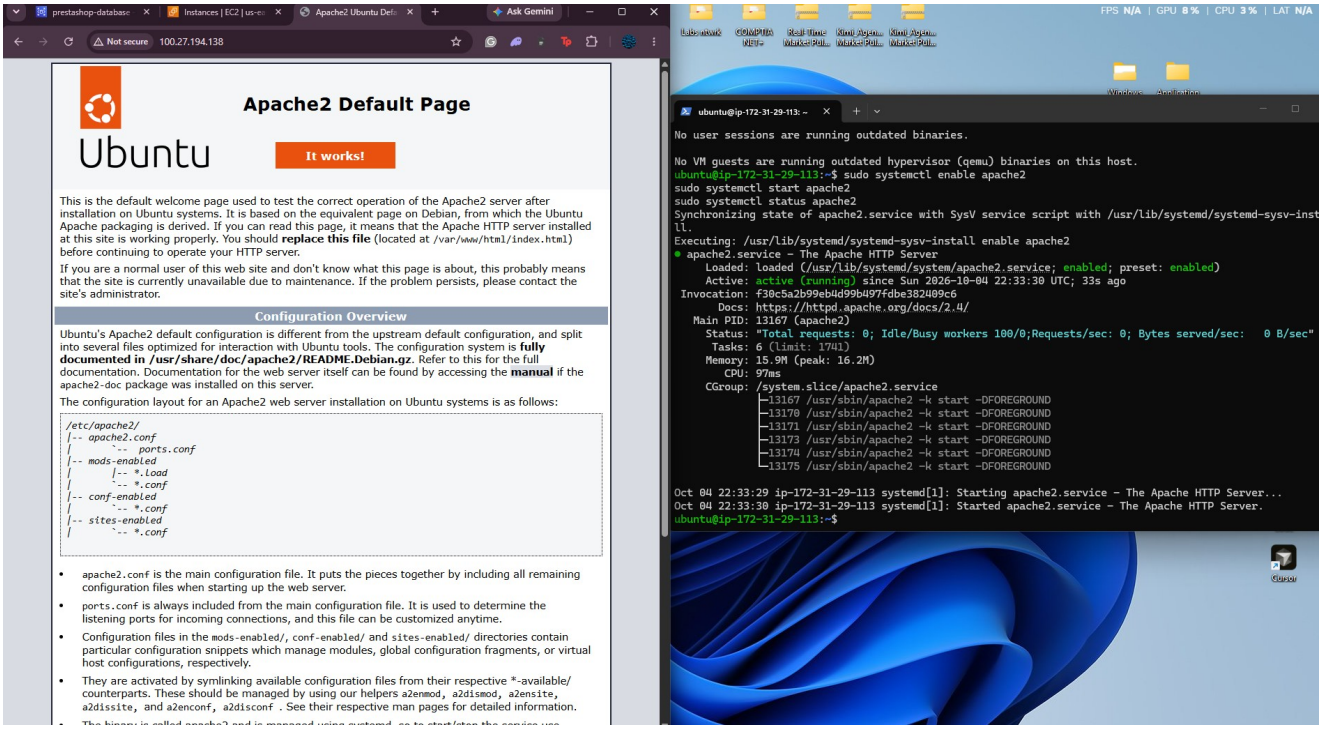


Figure 4. Apache running successfully and the default web page loading from the EC2 public IP.

6. PrestaShop Installation

I installed PrestaShop 9.2.0 on the EC2 server. The prepared Classic installer package was extracted into /var/www/html, ownership was assigned to the Apache web user, and file permissions were set before restarting Apache. The web-based PrestaShop Installation Assistant then became accessible through the EC2 public IP.

The store was configured with the demo products and recommended modules so the final public site would demonstrate a complete working installation.

7. Connecting PrestaShop to Amazon RDS

A database named prestashop was created on the RDS MySQL instance. In the PrestaShop installer, I replaced the local database address (127.0.0.1) with the RDS endpoint and used the RDS database credentials. The installer successfully tested the connection before installation continued.

Database server	prestashop-database.ckpi3acurtdm.us-east-1.rds.amazonaws.com
Database name	prestashop
Database user	prestashopadmin
Port	3306
Database password	Hidden / not included in documentation

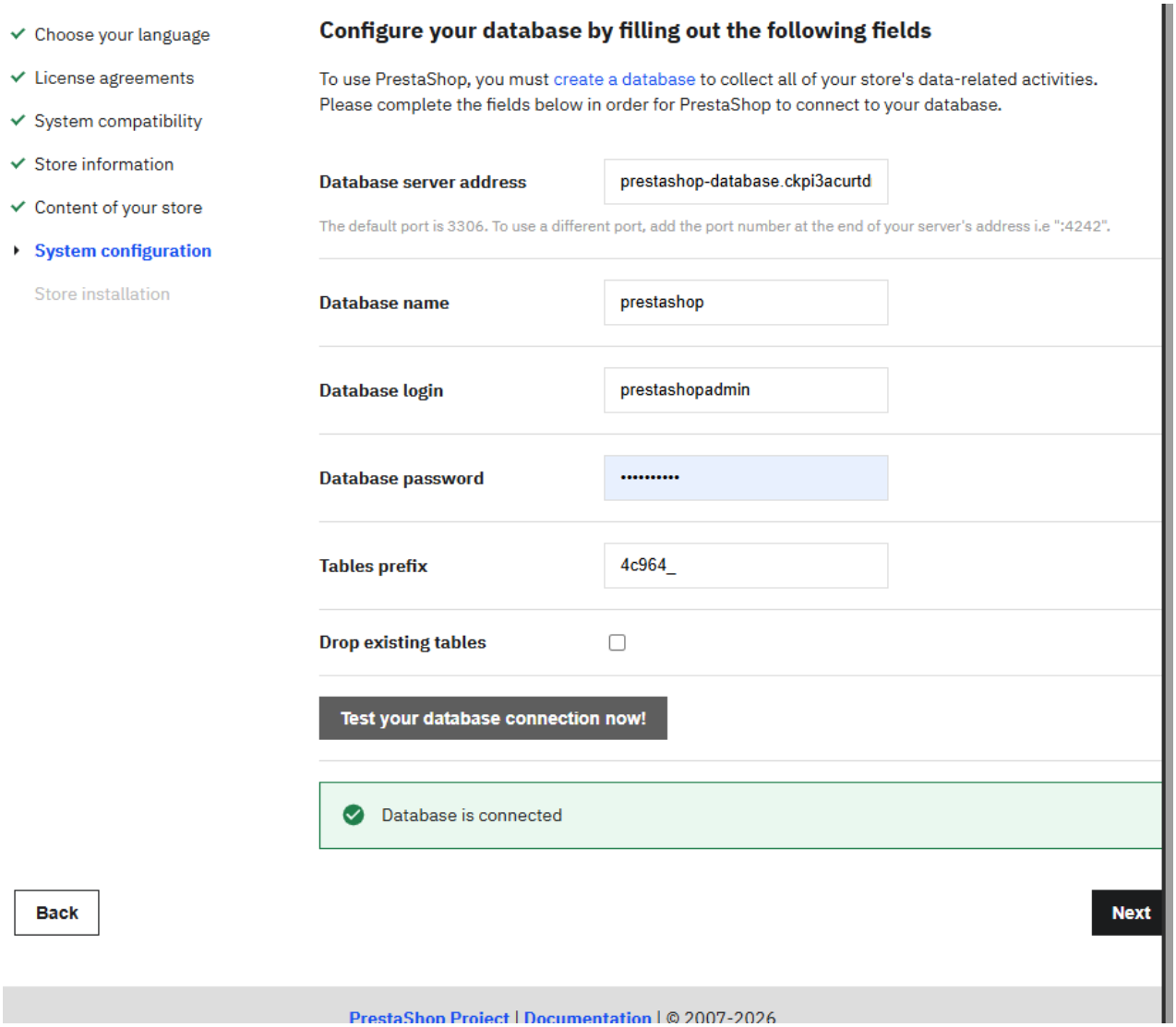


Figure 5. PrestaShop confirms that the separate RDS database is connected.

8. Installation Completion and Security Cleanup

The PrestaShop installer completed successfully. After installation, the install directory was removed from the web root as instructed by PrestaShop. Removing this folder prevents the installation process from being run again and is an important post-installation security step.

sudo rm -rf /var/www/html/install

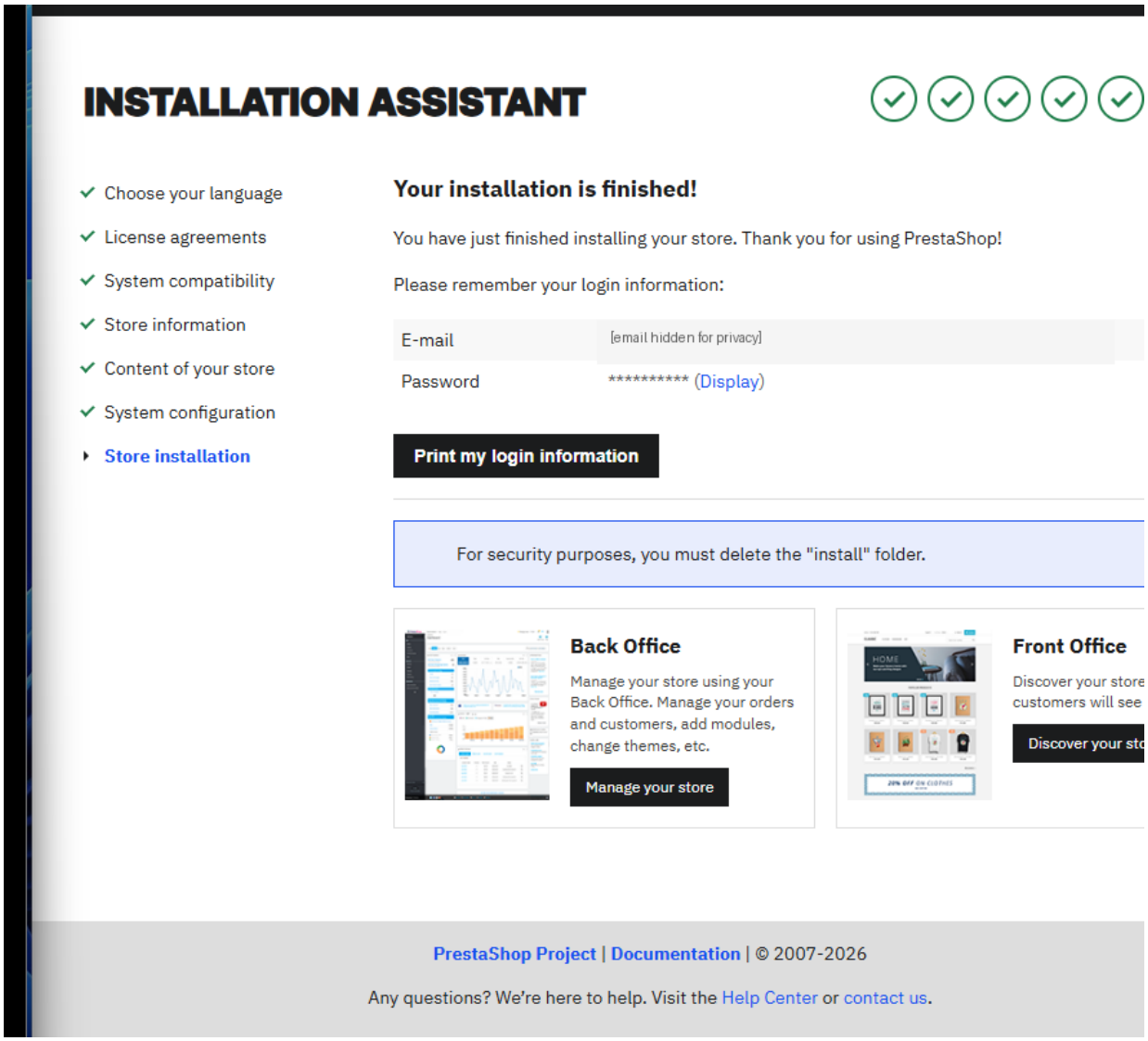


Figure 6. PrestaShop installation completed successfully. Login email is hidden in this report for privacy.

9. Final Working Storefront

The PrestaShop Front Office loaded successfully through the EC2 public IP. This confirms that the Apache/PHP application server, PrestaShop application, and RDS database are working together correctly.

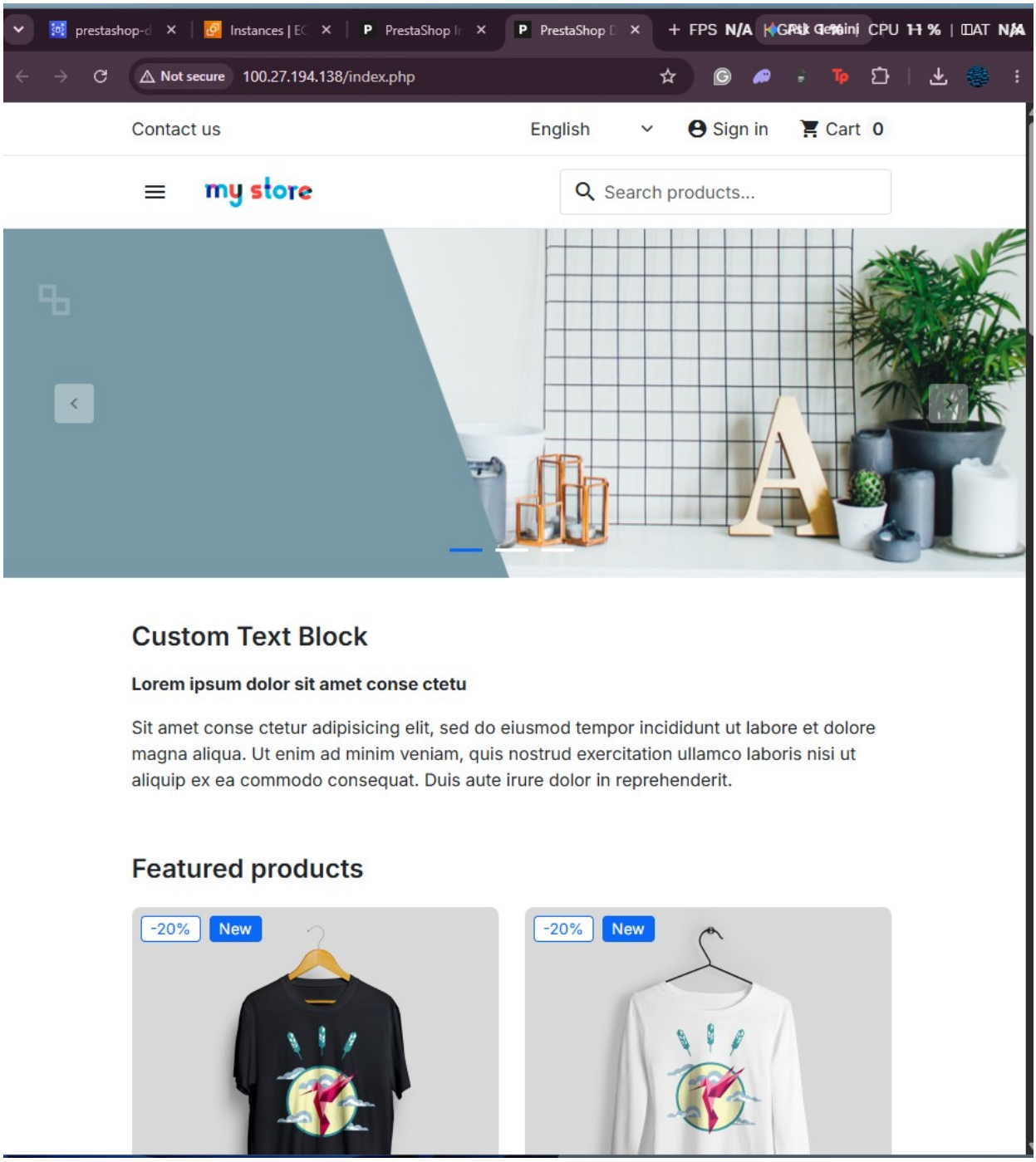


Figure 7. Final publicly accessible PrestaShop storefront.

10. Final Result

The assignment was completed successfully. PrestaShop is hosted on a new AWS EC2 instance, the MySQL database is hosted separately on Amazon RDS, and the storefront is publicly accessible. The database is not publicly exposed, and access to MySQL is restricted through the security group configuration.

Submitted URL: http://100.27.194.138/index.php

Configuration Summary

- Application server: Amazon EC2 (Ubuntu)
- Web server: Apache 2
- Runtime: PHP 8.5
- Application: PrestaShop 9.2.0
- Database server: Amazon RDS MySQL
- Database publicly accessible: No
- Public web ports: 80 and 443
- Database port: 3306 restricted to the PrestaShop security group